CRAFT Maker Space · Mary Immaculate College · companion to Code a Tiny Computer

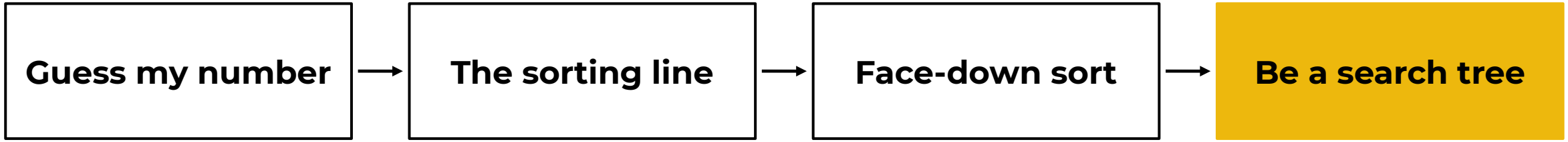
Sort and Search, Unplugged

Paper and movement activities for 5th and 6th class

Mike Borowczak, Ph.D.
Associate Professor of Electrical and Computer Engineering
University of Central Florida



Four activities, no screens



Search · 5 min | Sort · 10 min | Sort · 12 min | Search · 20 min

Your team of three

Runner

Moves or flips the cards

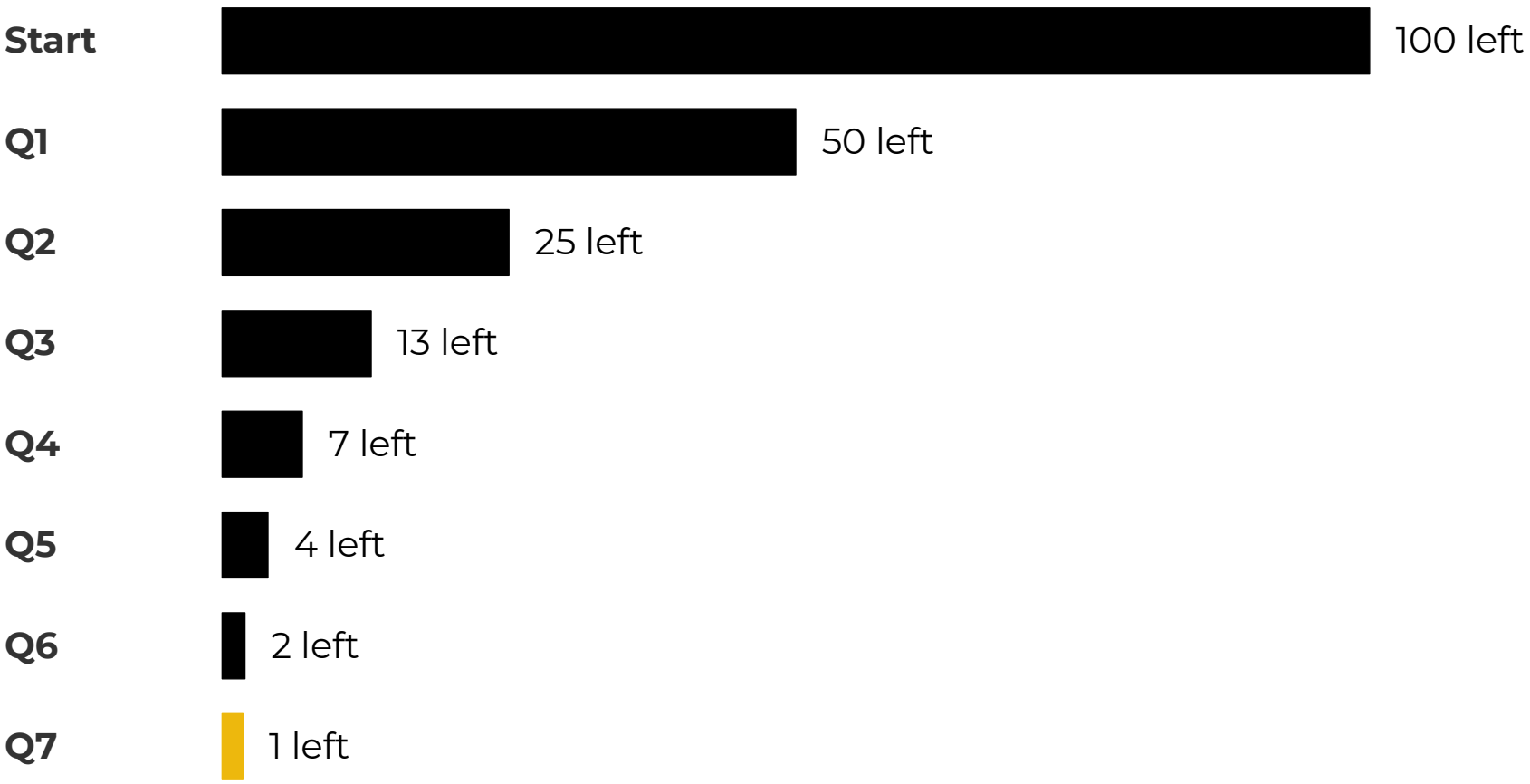
Tester

Checks every move follows the rule

Reporter

Keeps the count and tells the room

1 · Guess my number: 1 to 100



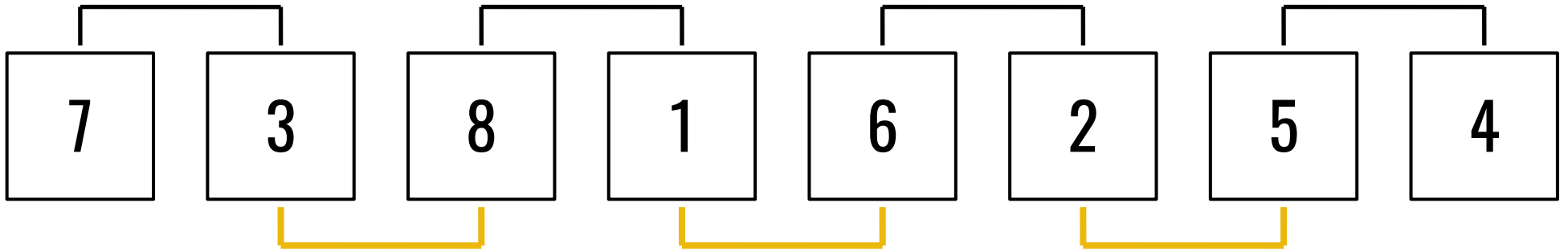
7

questions are always enough, if every question cuts what is left in half

Ask only: “Is it bigger than ___?”

2 · The sorting line

Odd claps: these partners



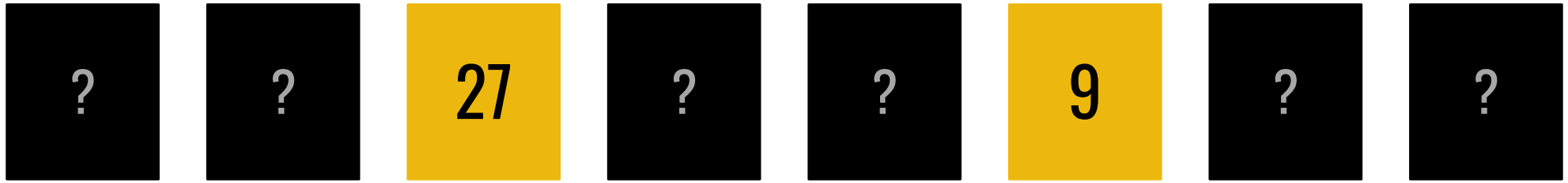
Even claps: these partners

- Eight pupils in a row, each holding a number card
- On each clap, partners compare: bigger number on the left? Swap
- Reporter counts claps until the whole row is in order

8

claps are always enough for 8 people

3 · Face-down sort: count every comparison



Turn over two cards, compare, turn them back: that is one comparison

Strategy A: smallest first

- Find the smallest card, move it to the start
- Repeat with the cards that are left

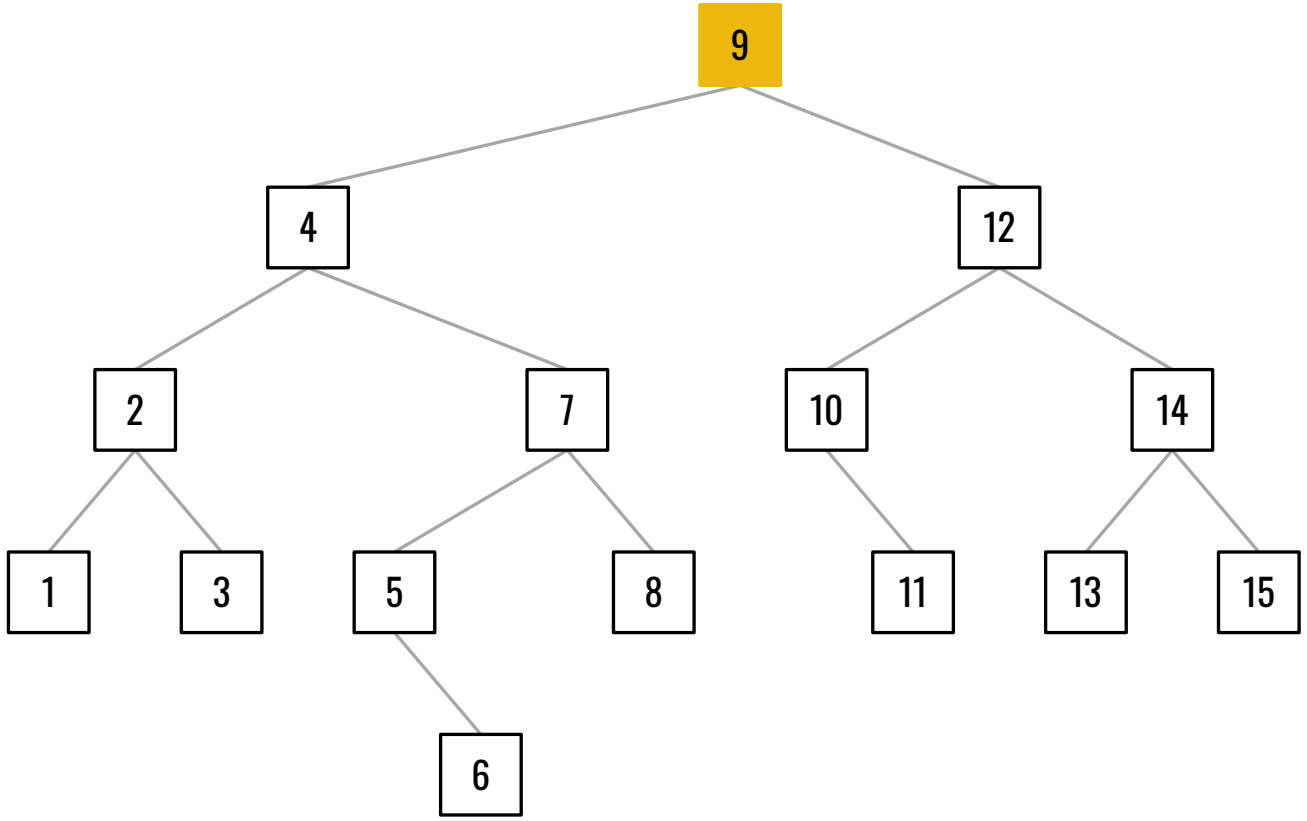
Strategy B: split and merge

- Split into two piles of four and sort each pile
- Merge: compare the front cards, take the smaller

Reporter tally each comparison. Which strategy needed fewer?

4 · Be a binary search tree

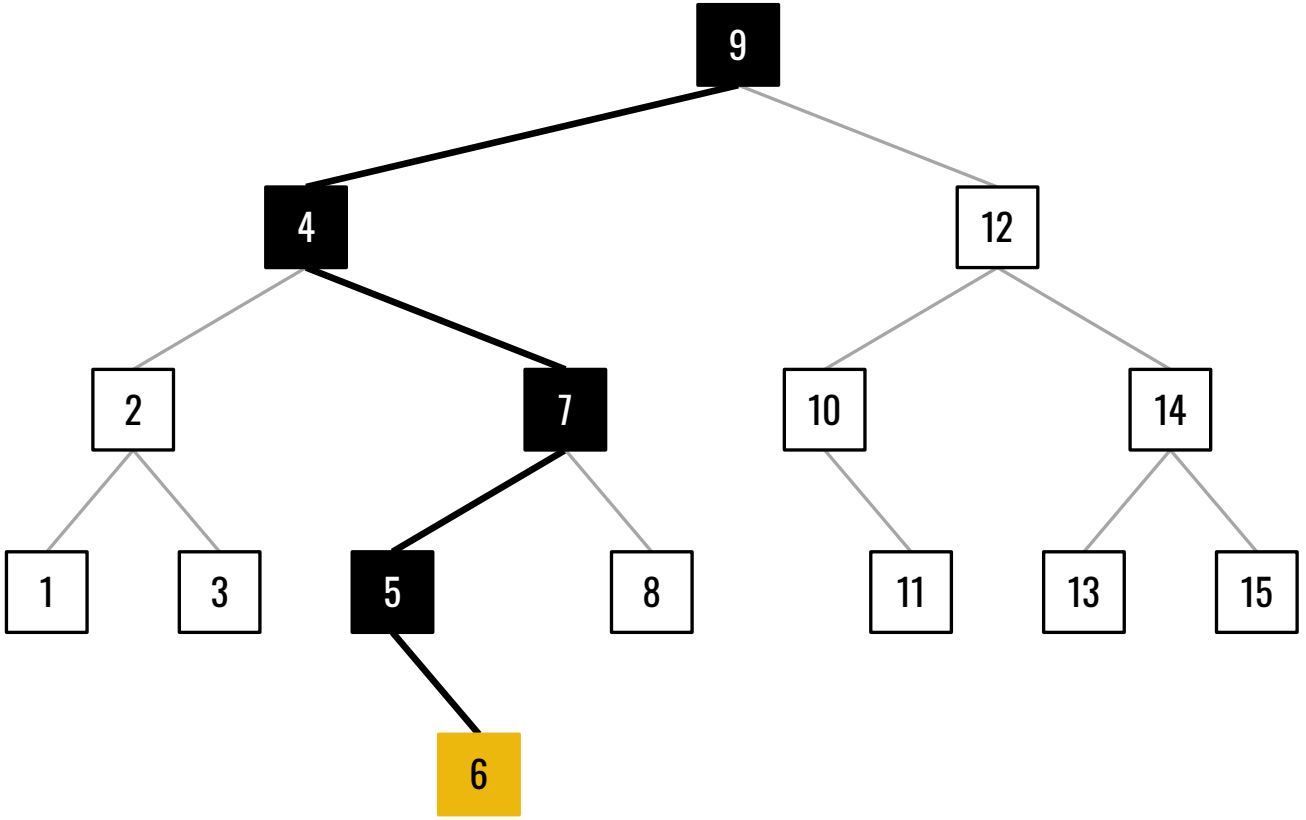
- The first pupil is the root and stands at the front
- Each new pupil starts at the root and asks: am I smaller or larger than you?
- Smaller: go to the SMALLER side.
  Larger: go to the LARGER side
- Ask the next person there; stop when nobody is there, and stand one step behind



← SMALLER **LARGER →**

Example: 15 pupils joined in this order: 9, 4, 12, 7, 2, 14, 10, 5, 1, 13, 15, 3, 8, 11, 6

Search the tree, then search a line



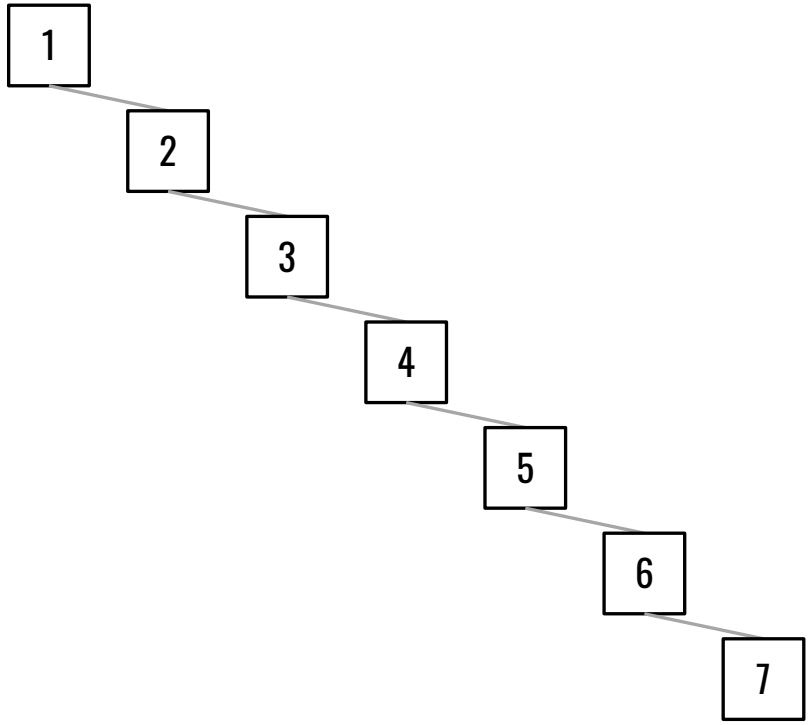
5 questions at most in this tree

15 questions at most in a line of the same 15 pupils

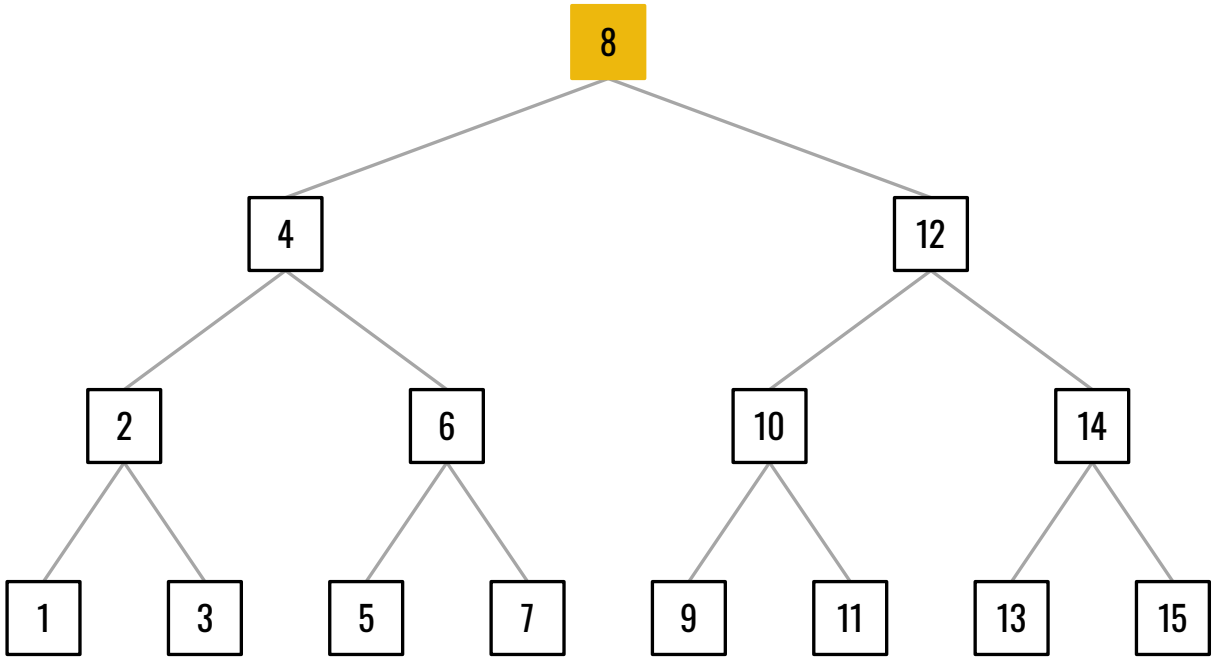
- A searcher gets a number to find, starts at the root, and asks only “smaller or larger?”
- Reporter counts the questions

Find 6: 9 → 4 → 7 → 5 → 6

Same numbers, different trees



... down to 15



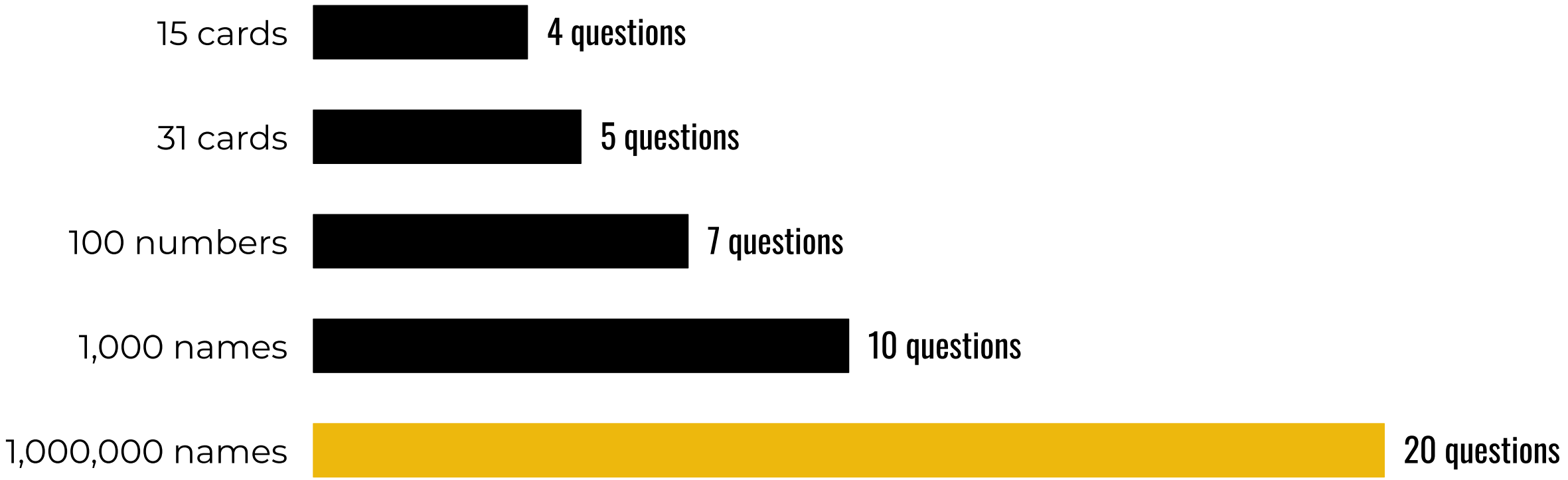
Joined in order 1, 2, 3 ... 15

A stick: up to 15 questions, as slow as a line

Joined in order 8, 4, 12, 2, 6, 10, 14, then the odd numbers

A bushy tree: 4 questions at most

Halving is powerful



Most questions needed when every question halves what is left

Back at the micro:bit: it guesses your number

on start	`low = 1 · high = 100 · call makeGuess`
function	`makeGuess: guess = floor((low + high) ÷ 2), show it`
button A	`smaller: high = guess − 1 · call makeGuess`
button B	`larger: low = guess + 1 · call makeGuess`
A + B	`found it: show a tick`

Tester

- Pick a secret number
- Press A or B after each guess
- Did it always find you in 7 guesses or fewer?

Extension for fast teams · build it in makecode.microbit.org with variables and the floor block from Math

Sorted things are quick to search. Every good question cuts the rest in half.

For the teacher: setup and materials

Activity	Time	You need
1 Guess my number	5 min	A slip of paper for the secret number
2 The sorting line	10 min	Number cards 1–31 (A5), one per pupil; open floor for rows of 8
3 Face-down sort	12 min	One set of 8 small cards per team; a Reporter sheet per team
4 Be a search tree	20 min	Cards 1–15; tape for a ROOT spot; SMALLER and LARGER wall signs; a clear floor about 6 m × 4 m
Extension	any	micro:bits and laptops from the main workshop

Printables: number cards, team card sets, wall signs, and the Reporter sheet (separate PDF) · inspired by CS Unplugged (csunplugged.org)